

CCT College Dublin

Assessment Cover Page

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Lecturer Name:	<i>Kelvin Martins</i>
Student Full Name:	Christopher Antonio Anich Escarpentier
Student Number:	2023202
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Research Proposal: Comparing Custom and Transfer-Learning CNN Models for Chest X-ray Classification: Evaluating Performance and Scalability with Kubernetes Orchestration

Name: Christopher Anich

Student no: 2023202

Course: M.Sc. In Data Analytics

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Affiliation: CCT College Dublin, Ireland

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1. Topic Area

The field of medical diagnostics has witnessed a substantial transformation through the integration of Artificial Intelligence (AI), especially Convolutional Neural Networks (CNNs), to automate and enhance the accuracy of radiological image analysis. Chest X-ray (CXR) imaging is a primary diagnostic tool frequently employed in the identification of respiratory diseases, including viral pneumonia, bacterial pneumonia, and notably COVID-19 (Obuchowicz *et al.*, 2025). The COVID-19 pandemic underscored significant limitations in traditional diagnostic methods such as RT-PCR, which, despite high specificity, often suffer from delayed results, resource-intensive processes, and variable sensitivity (Wang, Lin and Wong, 2020).

Recent advances in deep learning, specifically CNN-based architectures, have offered promising solutions for rapid, accurate, and automated diagnosis from CXR images. However, conventional CNN architectures, despite their high accuracy, frequently require significant computational resources, limiting their practical deployment, especially in resource-constrained environments. This limitation necessitates exploring innovative methods that balance computational efficiency with high diagnostic accuracy (Yen and Tsao, 2024).

In response to this challenge, lightweight CNNs have been developed to reduce computational complexity while maintaining strong classification performance. Yen and Tsao (2024) introduced a model that combined high accuracy with efficiency, demonstrating clinical applicability. Meanwhile, containerisation technologies like Kubernetes offer scalable and secure deployment for AI models, addressing resource allocation, reproducibility, and management challenges (Pääkkönen *et al.*, 2021).

This project aims to evaluate the effectiveness and practicality of custom lightweight CNNs compared to transfer learning models (e.g. MobileNetV2, EfficientNet-B0, ResNet-50) for CXR classification, focusing on accuracy, sensitivity, specificity, and computational efficiency. It also leverages Kubernetes orchestration to assess whether lightweight models can meet the operational demands of clinical deployment.

By conducting this comparative analysis, the research addresses a current and pressing gap within the intersection of medical diagnostics, deep learning, and cloud-native technologies, aiming to make a meaningful and original contribution that enhances rapid diagnostic capabilities, especially pertinent in scenarios similar to the COVID-19 pandemic. The outcomes of this research could significantly impact clinical workflows, improving diagnostic turnaround times, accuracy, and overall patient management, particularly in environments with constrained computational resources.

2. Research Objectives and Hypotheses.

The main objective of this research is to conduct a comparative data analytics evaluation of a custom lightweight convolutional neural network (CNN) and three widely used transfer learning models, MobileNetV2, EfficientNet-B0, and ResNet-50 (Mirzaee and Ghorbanzadeh, 2025), for classifying chest X-ray (CXR) images into three clinically relevant categories: normal lungs, community-acquired pneumonia (CAP), and pulmonary tuberculosis (TB), which remains a major global health concern (Chen *et al.*, 2024).

Beyond assessing classification accuracy, sensitivity, and specificity, the study also incorporates operational performance metrics, including inference latency and resource usage. These analytics will be gathered from a simulated Kubernetes environment to evaluate model efficiency under realistic deployment conditions, thereby supporting evidence-based selection for clinical use (de Almeida *et al.*, 2025).

Research Objectives

1. To design a small, efficient custom CNN model using lightweight architecture techniques (e.g. projection-expansion blocks) that can run effectively in constrained environments.
2. To build a balanced dataset from two open-access CXR sources, CheXpert for normal and pneumonia cases, and TBX11K for tuberculosis, combining around 5,000 images per class and pre-processed to 256×256 resolution for training.
3. To compare the performance of the custom CNN with MobileNetV2, EfficientNet-B0, and ResNet-50 using 5-fold cross-validation. Evaluation metrics will include accuracy, sensitivity, specificity, and confusion matrices.
4. To containerise the training and evaluation pipeline using Podman and deploy it in a simulated Kubernetes cluster (1 control-plane + 3 worker nodes) with MinIO storage. Resource usage and model loading speed will also be monitored.
5. To apply basic explainability methods, such as Grad-CAM, to qualitatively assess whether model predictions focus on clinically plausible image regions.

Hypotheses

- H1: There is no statistically significant difference ($p > 0.05$) in classification performance (accuracy, sensitivity, and specificity) between the custom lightweight CNN and the best-performing transfer learning model (based on mean F1-score), as evaluated across five cross-validation folds using paired t-tests or Wilcoxon signed-rank tests.
- H2: The containerised custom CNN model will show statistically lower mean inference latency and memory usage ($p < 0.05$) compared to MobileNetV2, EfficientNet-B0, and ResNet-50 under Kubernetes deployment, as measured via Prometheus metrics and compared using ANOVA or Kruskal-Wallis tests.

3. Literature Review.

3.1 Introduction

Deep learning has become increasingly influential in medical diagnostics, especially in image-based tasks such as chest X-ray (CXR) interpretation. Among these techniques, Convolutional Neural Networks (CNNs) have proven effective in detecting a range of pulmonary conditions, offering diagnostic support in situations where radiologist availability is limited (Obuchowicz *et al.*, 2025). In recent years, many studies have explored the potential of CNNs for classifying CXRs into conditions like pneumonia, tuberculosis, and COVID-19. While COVID-focused work is abundant, relatively fewer studies address community-acquired pneumonia (CAP) and pulmonary tuberculosis (TB), two of the most common and deadly infectious respiratory diseases globally (Wang, Lin and Wong, 2020).

This review focuses on research that informs the classification of normal, CAP, and TB CXR images using CNNs. It also discusses the benefits of lightweight models, the role of transfer learning, and the emerging

practice of deploying models through containerised systems like Kubernetes (Psomakelis *et al.*, 2023). Explainability and ethical concerns are also discussed, given their importance for clinical applications and real-world adoption (Lekadir *et al.*, 2025).

3.2 CNN Architectures for Chest X-ray Classification

CNNs have become central to image analysis tasks because of their ability to learn local patterns in pixel data. In CXR analysis, they allow systems to detect opacities, cavities, and other radiographic features associated with lung disease (Liu, Yu and Tan, 2024).

Several studies have developed CNN models tailored for chest X-ray (CXR) classification. One lightweight architecture distinguished COVID-19, pneumonia, and normal cases with low latency and minimal parameters, optimised for real-time inference in resource-constrained settings (Yen and Tsao, 2024). Its efficiency and accuracy make it ideal for edge deployments such as rural clinics and mobile diagnostic units, where rapid and reliable decision support is essential.

Transfer learning is widely used in medical imaging. Rather than training CNNs from scratch, models like ResNet, EfficientNet, and MobileNet are pre-trained on large datasets (e.g. ImageNet) and fine-tuned for medical tasks, improving performance and reducing training time. For instance, an ensemble of deep transfer models was used to enhance TB detection accuracy on CXR images (Kotei and Thirunavukarasu, 2022), demonstrating the continued effectiveness of transfer learning, particularly with limited datasets.

3.3 Classification of Normal, Pneumonia, and Tuberculosis

Distinguishing between normal lungs, pneumonia, and TB is both clinically important and technically challenging. Pneumonia and TB may show overlapping radiological features like infiltrates and consolidation, though TB often presents with upper-lobe cavitation and fibrosis. Accurate multi-class classification can aid triage, especially where molecular testing is limited (Majumder, 2025).

A lightweight CNN model was proposed to classify chest X-ray images into pneumonia, tuberculosis (TB), and healthy categories. The model achieved over 90% accuracy while maintaining efficiency in terms of inference time and memory consumption. Including both community-acquired pneumonia (CAP) and TB in the same classifier was argued to enhance real-world utility, particularly in low-resource settings (Bhandari *et al.*, 2022).

A four-class CNN model was developed to identify COVID-19, viral pneumonia, bacterial pneumonia, and normal lungs. Although the study did not focus on tuberculosis (TB), the model achieved an accuracy of 97.94%, illustrating the strength of custom CNN architectures in addressing multi-class medical classification tasks (Karthik, Menaka and M., 2021).

Together, these studies confirm that CNNs can successfully distinguish between several lung conditions, even when classes share subtle visual similarities. However, training data must be carefully curated to ensure clear class separation and reduce the risk of bias.

3.4 Kubernetes and Model Deployment in Healthcare

AI models are only as valuable as their ability to operate effectively in the field. A key challenge in medical AI is the safe, scalable deployment of trained models. Kubernetes, an open-source platform for managing containerised applications, has become a preferred solution for orchestrating machine learning workflows in production settings (Koneru, 2025).

A secure, scalable architecture for deploying AI models via Kubernetes has been detailed in the literature, highlighting best practices such as rootless containerisation, network isolation, role-based access control

(RBAC), and image signing. These measures are critical in healthcare, where security and compliance are paramount, even for indirect handling of sensitive patient data (Morić *et al.*, 2025).

Recent advancements show that deploying PyTorch models using TorchScript and TorchServe within Kubernetes supports efficient inference through containerisation. This approach simplifies deployment, enables traceability and performance monitoring, and supports robust autoscaling to maintain consistent inference times under varying workloads (Schroeder, Boehm and Lampe, 2024).

Beyond reliability, Kubernetes provides tools for performance evaluation under realistic conditions. Features like Horizontal Pod Autoscaling and resource quotas facilitate the assessment of key metrics such as response time, CPU/GPU utilisation, and throughput, factors that are critical in clinical settings (Schroeder, Boehm and Lampe, 2024).

3.5 Explainability and Interpretability in Medical AI

The success of AI in medicine depends not only on accuracy but also on interpretability. Clinicians are unlikely to trust a model's outputs unless they can understand and verify how decisions are made. Explainability tools like Grad-CAM (Gradient-weighted Class Activation Mapping) address this need by generating heatmaps that indicate which image regions influenced the prediction. For pneumonia or TB classification, these may highlight features such as consolidation, nodules, or apical cavities, helping clinicians confirm alignment with clinical reasoning.

Vision Transformers (ViTs) have recently emerged as an alternative to CNNs for medical image classification. Unlike CNNs, ViTs use self-attention to capture global image dependencies instead of relying on local feature extraction. When paired with Grad-CAM, they can produce interpretable heatmaps that aid clinical understanding. This method has been successfully applied to tuberculosis detection in chest X-rays, showing high diagnostic accuracy and meaningful visual explanations (Vanitha *et al.*, 2025). However, ViTs were excluded from this study due to their higher computational demands and less mature edge deployment tooling, which conflict with the project's focus on practical scalability.

While explainability techniques like Grad-CAM are now common in research, their integration into production pipelines remains limited. Embedding such tools into the standard inference process, rather than treating them as optional, post-hoc analyses, would improve transparency and foster greater trust in clinical deployment (Vanitha *et al.*, 2025).

3.6 Ethical Considerations in Model Development and Deployment

Using AI in medical imaging raises several ethical challenges. Data privacy is paramount, especially in systems connected to electronic medical records or cloud storage. Fortunately, most datasets used in this domain, including CheXpert v1.0 and TBX11K, are anonymised and approved for research use (Parra-Ullauri *et al.*, 2023).

Bias is another concern. If the training dataset is imbalanced by age, gender, or demographic distribution, the model might perform better on some groups than others. Many authors, including Albahar *et al.* (2024), stress the need for fairness audits and diverse datasets to mitigate this risk. In this project, efforts will be made to maintain balance between classes and monitor per-class performance to avoid such problems (Patel *et al.*, 2024).

Finally, deployment in clinical environments requires more than just technical success. Compliance with local data protection laws (e.g. GDPR in the EU), hospital IT protocols, and institutional ethical approvals are all essential steps. Kubernetes offers several tools to support these requirements, but human oversight remains key (Boda and Immaneni, 2021).

3.7 Conclusion

The use of CNNs for CXR classification has advanced rapidly in recent years. While large transfer learning models still offer strong performance, lightweight custom CNNs are becoming increasingly practical and competitive, especially when deployed in containerised environments like Kubernetes. Research shows that models can now distinguish between normal lungs, pneumonia, and TB with high accuracy, particularly when explainability tools like Grad-CAM are integrated into the workflow.

However, technical success alone is not sufficient. Ethical deployment, fairness across demographics, and transparency in decision-making must also be addressed. This review confirms that the proposed project, a comparative study of lightweight and transfer learning CNNs deployed via Kubernetes for classifying normal, CAP, and TB CXR images, is relevant, realistic, and well-supported by current research.

4. Proposed Primary Research Methodology.

4.1 Research Design Overview

This research adopts a quantitative, experimental design to compare the performance and scalability of a custom lightweight Convolutional Neural Network (CNN) with three widely used transfer learning models: MobileNetV2, EfficientNet-B0, and ResNet-50. These models will be trained and evaluated to classify chest X-ray (CXR) images into three categories: normal lungs, community-acquired pneumonia (CAP), and pulmonary tuberculosis (TB). The methodology covers data collection, preprocessing, model development, evaluation, and Kubernetes-based deployment to assess not only accuracy but also practical feasibility in containerised environments.

4.2 Data Acquisition and Preprocessing

To build a balanced and representative dataset, publicly available and de-identified sources will be used. CheXpert v1.0 from the Stanford ML Group dataset will provide high-quality images for both normal and pneumonia cases (Irvin *et al.*, 2019). For tuberculosis, the study will rely on the TBX11K dataset, which includes more than 11,000 annotated CXR images (Liu *et al.*, 2020). These two datasets offer sufficient diversity and volume to support multi-class classification while ensuring ethical data usage.

All images will be resized to 256×256 pixels, balancing the trade-off between preserving anatomical detail and minimising computational load. Normalisation will be applied to scale pixel values between 0 and 1, standardising inputs for neural network training. To improve generalisation and mitigate class imbalance, the dataset will be augmented with common transformations such as horizontal flipping, random rotations, and zoom operations. These steps ensure model robustness without increasing the original dataset size excessively (Lin *et al.*, 2021).

4.3 Model Development

The proposed lightweight CNN will be developed from scratch with a focus on efficiency, portability, and compatibility with constrained environments. It will use depthwise separable convolutions to reduce parameter count, with batch normalisation and dropout to improve training stability and reduce overfitting. ReLU will serve as the activation function (Yen and Tsao, 2024).

Three pre-trained CNNs, MobileNetV2, EfficientNet-B0, and ResNet-50, will be fine-tuned using transfer learning, selected for their success in medical imaging and availability in standard deep learning libraries.

Their final classification layers will be replaced to support the three target classes: normal lungs, pneumonia, and tuberculosis. Initially, earlier layers will remain frozen to preserve general features, followed by gradual unfreezing for domain-specific fine-tuning. This approach leverages pre-training on large datasets like ImageNet, which is especially beneficial with limited labelled medical data (Mirzaee and Ghorbanzadeh, 2025).

4.4 Model Training and Evaluation

Training will be conducted following a structured and replicable protocol. The dataset will be partitioned into training (70%), validation (15%), and test (15%) sets, ensuring class balance and enabling fair and consistent evaluation (Patel *et al.*, 2024). The Adam optimizer will be employed with an initial learning rate of 0.001, selected to balance convergence speed and training stability. Categorical cross-entropy will serve as the loss function, being appropriate for multi-class classification tasks by measuring divergence between predicted and true label distributions.

To reduce overfitting and computation, early stopping will monitor validation loss and halt training once performance plateaus. Evaluation will include overall accuracy, class-wise precision, recall, and F1-score, with confusion matrices to visualise predictions across the three classes. ROC curves and AUC scores will be generated for each class to assess discriminatory ability. These procedures, based on Patel *et al.* (2024), will be applied uniformly to both the custom CNN and transfer learning models to ensure fair, transparent, and reproducible comparisons.

Before training, exploratory data analysis (EDA) will assess class distribution, resolution consistency, and dataset balance. Representative images from each class will also be reviewed to confirm data quality. To evaluate Hypothesis 1 (H1), model performance differences will be statistically tested using five-fold cross-validation results. For pairwise comparisons (e.g. custom CNN vs MobileNetV2), paired t-tests will be used for accuracy, sensitivity, specificity, and AUC if normality holds; otherwise, Wilcoxon signed-rank tests will be applied. For multiple comparisons, one-way ANOVA or Kruskal–Wallis tests will be used, with Bonferroni correction for post hoc analysis. A significance threshold of $p < 0.05$ will support evidence-based conclusions.

4.5 Operational Metric Collection Environment

A central objective of this research is to evaluate the operational performance of each model under simulated resource-constrained conditions. All models will be implemented using PyTorch and containerised with Podman to ensure portability and consistency. Each container will integrate TorchServe to support standardised inference execution across models (PyTorch, no date).

The runtime evaluation will take place in a local Kubernetes environment using Minikube, configured with one control-plane node and three worker nodes. This environment serves as a controlled simulation to measure how models perform under realistic deployment constraints. CPU usage, memory consumption, and inference latency will be monitored during inference operations using built-in Kubernetes resource tracking and custom logging.

If needed, MinIO, a lightweight, self-hosted object storage system, will emulate typical data access patterns. Prometheus will collect system metrics, and Grafana will visualise performance over time. To evaluate Hypothesis 2 (H2), statistical tests will assess whether the custom CNN achieves significantly lower mean latency and memory usage than the transfer learning models, based on Prometheus-monitored inference metrics. This controlled environment enables the collection of analytic data to compare runtime efficiency and support evidence-based conclusions on deployment practicality. (Liu, 2019).

4.6 Explainability and Interpretability

In clinical applications, black-box models without interpretability are rarely accepted. To address this, the study will incorporate Grad-CAM (Gradient-weighted Class Activation Mapping) to generate visual explanations for model predictions. Grad-CAM helps visualise the regions of an input image that most strongly influence the decision, offering transparency and interpretability that clinicians can understand. This step is not just optional; it is essential for clinical trust, even in research or pilot-stage deployments (Patel *et al.*, 2024).

Grad-CAM heatmaps will be generated at inference time by embedding visualisation logic into custom TorchServe handlers. These will be served alongside classification outputs via the API and logged as image artefacts for post hoc review. To assess interpretability, the heatmaps will be qualitatively evaluated by checking alignment with clinically recognised features, such as apical cavitations in TB and lower-lobe consolidations in pneumonia (Devasia *et al.*, 2023). This alignment supports both technical validation and clinical interpretability, essential for responsible AI deployment in medical imaging.

4.7 Ethical Considerations

Although ethical aspects will be detailed in Section 5, it is important to emphasise that all data used in this project is publicly available and anonymised. The research will not involve any direct human participation or clinical trials. Still, all data handling and model deployment steps will follow ethical best practices and be guided by principles of transparency, privacy, and fairness.

5. Ethical Considerations.

5.1 Data Privacy and Patient Confidentiality

The datasets utilized in this study, including the CheXpert v1.0 from the Stanford ML Group (Irvin *et al.*, 2019) and TBX11K (Liu *et al.*, 2020), are publicly available and de-identified, ensuring that no personally identifiable information (PII) is present. This adherence to data privacy standards mitigates concerns related to patient confidentiality and complies with ethical research practices. Nevertheless, it is imperative to handle all data with the utmost care, ensuring secure storage and access controls to prevent unauthorized usage.

5.2 Informed Consent and Data Usage

Given that the datasets are publicly available and de-identified, explicit informed consent from patients is not required for this research. However, it is essential to acknowledge and respect the original data collection protocols, which would have included obtaining consent and ethical approvals. Proper citation and acknowledgment of the data sources will be maintained throughout the study to honour the contributions of the original data providers (Avanzato *et al.*, 2024).

5.3 Bias and Fairness in Model Development

Bias in machine learning can result from imbalanced datasets, causing uneven performance across demographic groups. To address this, the study will use data augmentation and ensure balanced representation across the three classes: normal lungs, CAP, and TB. Continuous monitoring will help detect and mitigate bias, supporting fair and equitable model performance (Yen and Tsao, 2024).

5.4 Explainability and Transparency

The deployment of AI models in clinical settings necessitates transparency and interpretability to foster trust among healthcare professionals. As discussed in Section 4.6, this study employs explainability tools such as Grad-CAM to highlight the regions of the image contributing to each prediction. The inclusion of these visual explanations is essential for clinical adoption, as it allows practitioners to verify that model outputs align with radiological reasoning (Patel *et al.*, 2024).

5.5 Deployment and Clinical Integration

Deploying AI models in healthcare environments introduces ethical considerations related to reliability, accountability, and patient safety. The study's deployment strategy, utilizing Kubernetes for scalable and efficient model serving, will include rigorous testing and validation to ensure robustness. Additionally, clear

documentation and user guidelines will be provided to assist clinicians in integrating the AI tool into their workflow responsibly (Avanzato *et al.*, 2024).

5.6 Compliance with Ethical Standards

The research will adhere to ethical standards and relevant legal frameworks, including IRB approval, GDPR, and the Declaration of Helsinki. In the Irish context, it will also comply with the Health Research Regulations 2018 and guidance from the Data Protection Commission (DPC). The forthcoming EU AI Act, particularly its provisions on high-risk healthcare systems, will be considered. Regular audits and reviews will ensure ongoing compliance and timely response to emerging ethical or regulatory concerns (Zeydan, Arslan and Liyanage, 2024).

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